Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology**	
Subject: DSIP	**Aim: open ended problem:-1**	
	Date:16-07-2026	**Enrollment No:92400133057**

Question:-1

Code:-

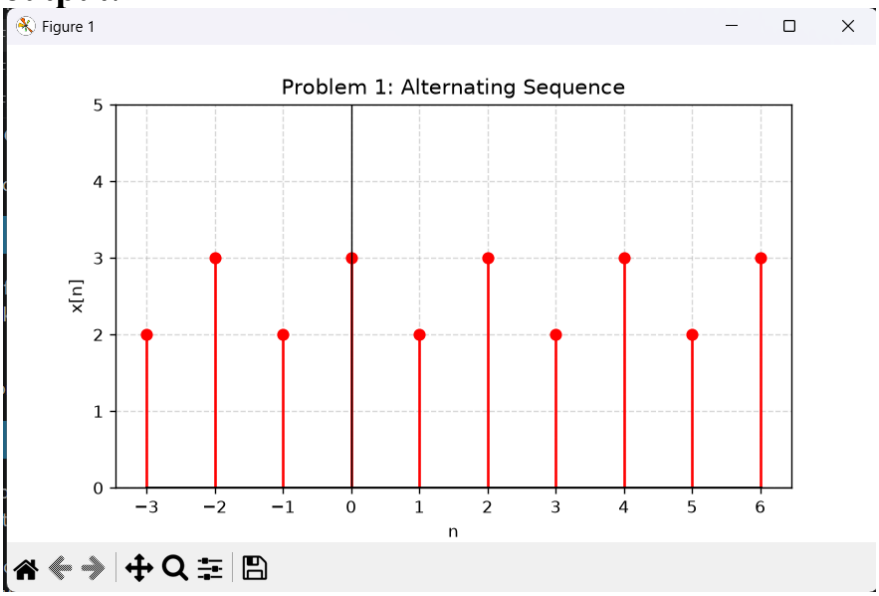
```
import numpy as np
import matplotlib.pyplot as plt

# Define time axis and amplitude values
n = np.arange(-3, 7)
# Alternating pattern: 2, 3, 2, Peak(3), 2, 3, 2, 3, 2, 3
x = np.array([2, 3, 2, 3, 2, 3, 2, 3, 2, 3])

# Plotting
plt.figure(figsize=(7, 4))
plt.stem(n, x, linefmt='r-', markerfmt='ro', basefmt='k-')
plt.title('Problem 1: Alternating Sequence')
plt.xlabel('n')
plt.ylabel('x[n]')
plt.xticks(n)
plt.ylim(0, 5)
plt.axhline(0, color='black', linewidth=0.8)
plt.axvline(0, color='black', linewidth=0.8)
plt.grid(True, linestyle='--', alpha=0.5)
plt.show()
```

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output:-



Question:-2

Code:-

```
import numpy as np

import matplotlib.pyplot as plt

# We add extra boundary points to make the block drop down cleanly to 0

n = np.array([-1, 0, 1, 2, 3, 4, 5])

x = np.array([0, 1, 2, 3, 2, 1, 0])

plt.figure(figsize=(7, 4))

# 'where=post' creates the perfect staircase look

plt.step(n, x, where='post', color='red', linewidth=2)

plt.title('Problem 2: Staircase Waveform')

plt.xlabel('n')
```

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```
plt.ylabel('x[n]')

plt.xticks(n)

plt.ylim(0, 4)

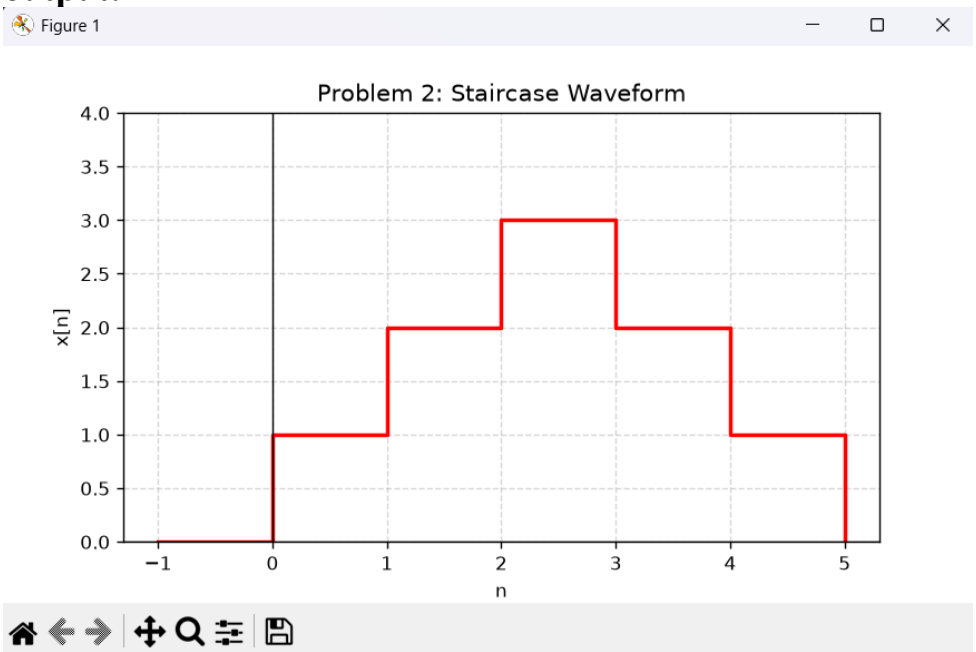
plt.axhline(0, color='black', linewidth=0.8)

plt.axvline(0, color='black', linewidth=0.8)

plt.grid(True, linestyle='--', alpha=0.5)

plt.show()
```

output:-



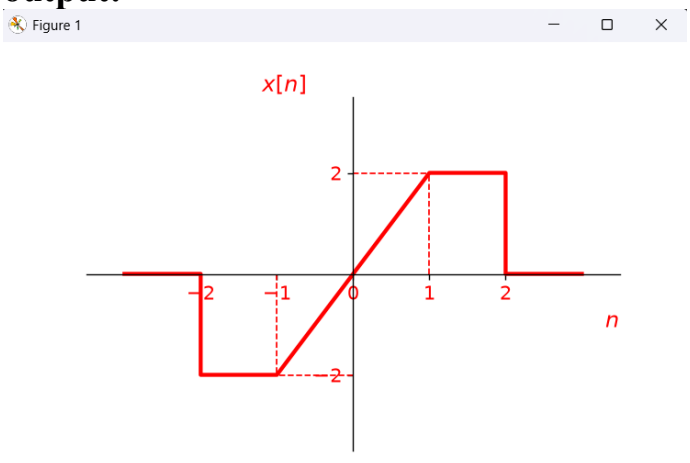
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	Date:16-07-2026	**Enrollment No:92400133057**

Question:-3

Code:-

```
import matplotlib.pyplot as plt
# Define the coordinates for the piecewise continuous signal
n = [-3, -2, -2, -1, 1, 2, 2, 3]
x = [0, 0, -2, -2, 2, 2, 0, 0]
plt.figure(figsize=(6, 4))
# Plot the main signal line in red to match the handwritten style
plt.plot(n, x, color='red', linewidth=2.5)
# Add dashed alignment lines
plt.plot([1, 1], [0, 2], color='red', linestyle='--', linewidth=1)
plt.plot([0, 1], [2, 2], color='red', linestyle='--', linewidth=1)
plt.plot([-1, -1], [0, -2], color='red', linestyle='--', linewidth=1)
plt.plot([0, -1], [-2, -2], color='red', linestyle='--', linewidth=1)
# Center the axes at the origin (0,0)
ax = plt.gca()
ax.spines['left'].set_position('zero')
ax.spines['bottom'].set_position('zero')
ax.spines['right'].set_color('none')
ax.spines['top'].set_color('none')
# Add labels
plt.xlabel('$n$', fontsize=14, color='red', loc='right')
plt.ylabel('$x[n]$', fontsize=14, color='red', loc='top', rotation=0)
# Set specific ticks matching the image
plt.xticks([-2, -1, 0, 1, 2], color='red', fontsize=12)
plt.yticks([-2, 2], color='red', fontsize=12)
# Set limits to frame the graph properly
plt.xlim(-3.5, 3.5)
plt.ylim(-3.5, 3.5)
plt.show()
```

output:-



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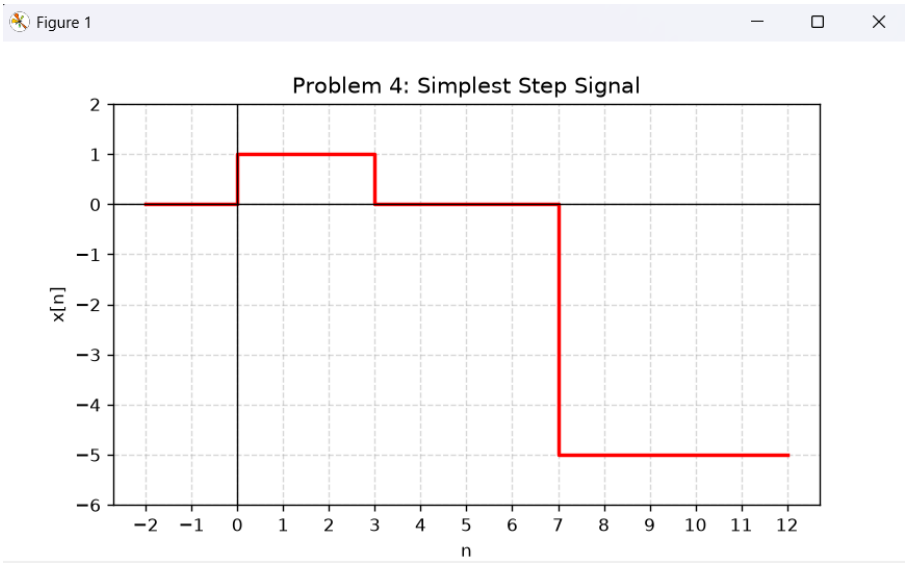
Question:-4

Code:-

```
import numpy as np
import matplotlib.pyplot as plt

# 1. Define the time values from -2 to 12
n = np.array([-2, -1,  0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12])
# 2. Directly type out the values we calculated for each point
# n=-2 to -1 are 0 | n=0 to 2 are 1 | n=3 to 6 are 0 | n=7 to 12 are -5
x = np.array([ 0,  0,  1,  1,  1,  0,  0,  0,  0, -5, -5, -5, -5, -5, -5])
# 3. Plot the step chart
plt.figure(figsize=(7, 4))
plt.step(n, x, where='post', color='red', linewidth=2)
# Formatting the chart
plt.title('Problem 4: Simplest Step Signal')
plt.xlabel('n')
plt.ylabel('x[n]')
plt.xticks(n)
plt.ylim(-6, 2)
plt.axhline(0, color='black', linewidth=0.8)
plt.axvline(0, color='black', linewidth=0.8)
plt.grid(True, linestyle='--', alpha=0.5)
plt.show()
```

output:-



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Question:-5

Code:-

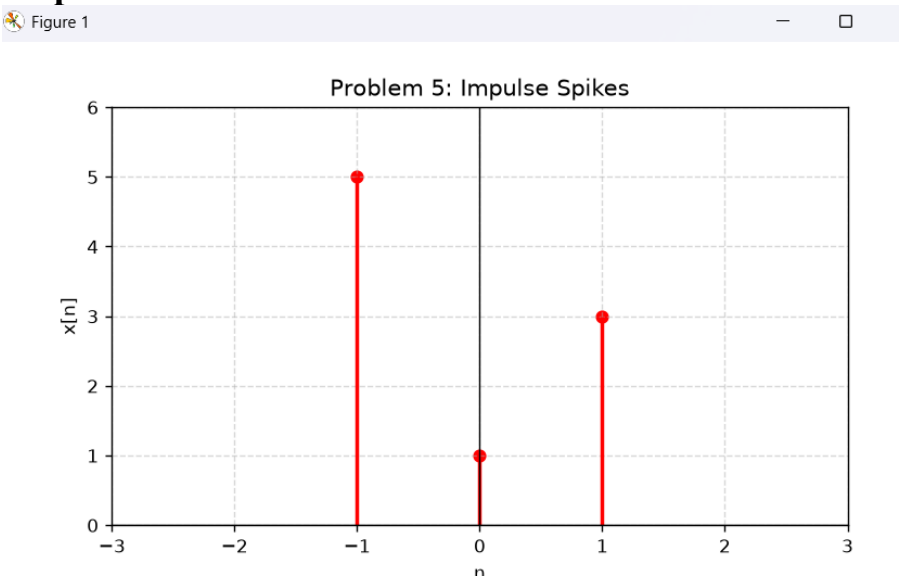
```
import numpy as np
import matplotlib.pyplot as plt

n = np.array([-1, 0, 1])
heights = np.array([5, 1, 3])  # 5 at n=-1, 1 at n=0, 3 at n=1

plt.figure(figsize=(7, 4))
# Draws sharp vertical lines from the 0 axis up to the height
plt.vlines(n, ymin=0, ymax=heights, colors='red', linewidth=2)
plt.scatter(n, heights, color='red') # Adds a dot at the top of each spike

plt.title('Problem 5: Impulse Spikes')
plt.xlabel('n')
plt.ylabel('x[n]')
plt.xlim(-3, 3)
plt.ylim(0, 6)
plt.axhline(0, color='black', linewidth=0.8)
plt.axvline(0, color='black', linewidth=0.8)
plt.grid(True, linestyle='--', alpha=0.5)
plt.show()
```

output:-



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